

Lesson 5

Random Variables and Their CDFs

It is not always easy to work
with Sample Spaces like

$\{H, T\}^n$, especially when we want
to quantify the properties of
such sample spaces. Therefore,
we work with a Random
Variable.

Loosely speaking, a random variable is a function from the sample space to real numbers.

Def. Assume that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space. The function $X: \Omega \rightarrow \mathbb{R}$ is a random variable if $\forall c \in \mathbb{R}, X^{-1}([-\infty, c])$ is $\mathcal{F}$ -measurable, i.e. it is an event,

In other words

$$\mathcal{X}^{\leftarrow}((-\infty, c]) = \left\{ \omega \in \Omega \mid \mathcal{X}(\omega) \in (-\infty, c] \right\} \in \mathcal{F}$$

is an event for all real numbers $c \in \mathbb{R}$.

Example: Consider $(\Omega, \mathcal{F}, \mathbb{P})$

where $\Omega = \{\alpha, \beta, \gamma\}$

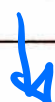
σ -field generated by $\{\alpha, \beta\}$
 $\mathcal{F} = \{\emptyset, \{\alpha, \beta\}, \{\gamma\}, \Omega\}$

and $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$

$\mathbb{P}(\{\alpha, \beta\}) = 1/2$, $\mathbb{P}(\{\gamma\}) = 1/2$

Consider $g: \Omega \rightarrow \mathbb{R}$

$$g(\omega) = \begin{cases} 1 & \omega \in \{\alpha, \beta\} \\ 2 & \omega \in \{\gamma\} \end{cases}$$



$$P(g(\omega) = 1) = P(\{\omega \mid g(\omega) \in \{\alpha, \beta\}\})$$

$$= P(\{\alpha, \beta\}) = \frac{1}{2}$$

$$P(g(\omega) = 2) = P(\{\omega \mid \omega \in \{\gamma\}\})$$

$$= P(\{\gamma\}) = \frac{1}{2}$$

$$P(g(\omega) \leq c) = \begin{cases} P(\emptyset) = 0 & c < 1 \\ P(\{\alpha, \beta\}) = \frac{1}{2} & 1 \leq c < 2 \\ P(\{\alpha, \beta, \gamma\}) = 1 & c \geq 2 \end{cases}$$

$\nwarrow$
 $g((-\infty, c])$

For all values of c , $P(\cdot)$ is well-defined / ~~not defined~~

Next, consider $h: \Omega \rightarrow \mathbb{R}$

$$h(\omega) = \begin{cases} 0 & \omega = \alpha \\ 1 & \omega = \beta \\ 2 & \omega = \gamma \end{cases}$$

$$\mathbb{P}(h(\omega) = 0) = \mathbb{P}(\{\omega \mid \omega = \alpha\})$$

$$= \mathbb{P}(\{\alpha\}) = \text{undefined}$$

$\notin \mathcal{F}$

Issue: The σ -field on which $\mathbb{P}$ is defined is not rich / large enough.

Convention: Upper-case letters
 are used for r.v.'s: X, Y, Z ,
 and lower case letters
 are used for numerical
 values that they can

take.

$$X(\omega) = x \in \mathbb{R}$$

$$Y(\omega) = y \in \mathbb{R}$$

It is an appropriate practice
 to show the dependence of
 a random variable on $\omega \in \Omega$:
 $X(\omega), Y(\omega), Z(\omega)$

Example : Indicator Function
of A Measurable Set.

Let $(\Omega, \mathcal{F})$ be a measurable
space and $A \in \mathcal{F}$.

The indicator function of A:

$$I_A : \Omega \rightarrow \{0, 1\}$$

$$I_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \quad \omega \in A^c \end{cases}$$

is a random variable.

$$\begin{aligned} X^{\leftarrow}((-\infty, 2]) &= \{\omega \in \Omega \mid X(\omega) \in (-\infty, 2]\} \\ &= \{\omega \in \Omega \mid X(\omega) \leq 2\} = \Omega \in \mathcal{F} \end{aligned}$$

$$X^{\leftarrow}((-\infty, -2]) = \{\omega \in \Omega \mid I_A(\omega) \leq -2\} = \emptyset \in \mathcal{F}$$

$$X^{\leftarrow}((-\infty, 1/2]) = \{\omega \in \Omega \mid I_A(\omega) \leq 1/2\} \\ = A^c \in \mathcal{F}$$

In general

$$X^{\leftarrow}((-\infty, c]) = \begin{cases} \emptyset & c < 0 \\ A^c & 0 \leq c < 1 \\ \Omega & c \geq 1 \end{cases}$$

Note: It is common to

show the event

$$\underline{X^{\leftarrow}((-\infty, 2])} \text{ as}$$

$$\{X \leq 2\} = \{\omega \in \Omega \mid X(\omega) \leq 2\}$$

↑
event

Example: $\mathcal{H}(\Omega, \mathcal{F})$ is a measurable space equipped with probability measure $\mathbb{P}$, and $A \notin \mathcal{F}$, I_A , the indicator function of A , is NOT a random

variable.

$$A \notin \mathcal{F}$$

$$I_A: \Omega \rightarrow \{0, 1\}$$

$$I_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$

$$\{X \leq 1/2\} = X^{-1}((-\infty, 1/2])$$

$$= \{\omega \mid X(\omega) \leq 1/2\} = \{\omega \mid I_A(\omega) \leq 1/2\}$$

$$A^c \in \mathcal{F} \Rightarrow (A^c)^c \in \mathcal{F} \Rightarrow A \in \mathcal{F} \text{ contradiction}$$

When $A \in \mathcal{F}$, I_A is called
a "Bernoulli Random Variable"

Bernoulli Random Variable
represents an experiment
that has two types of

outcome : success and failure

If A occurs (outcome belongs
to A), success occurred

If A^c occurs (outcome ^{doesn't} belongs
to A), failure occurred.

$$P(\text{success}) = P(X=1)$$

$$= P(I_A=1) = P(\{\omega \mid I_A(\omega)=1\})$$

$$= P(A)$$

$$P(\text{failure}) = P(X=0)$$

$$= P(I_A=0) = P(\{\omega \mid I_A(\omega)=0\})$$

$$= P(A^c) = 1 - P(A)$$

Example: A function of a r.v.

Assume that X is a r.v.

on $(\Omega, \mathcal{F})$. Also assume

that $Y: \Omega \rightarrow \mathbb{R}$

$$Y(\omega) = X^3(\omega)$$

Is Y a random variable?

In order to show that Y is

a random variable, we must

show that

$$\begin{aligned} \forall c \quad Y^{-1}((-\infty, c]) &= \{\omega \mid Y(\omega) \leq c\} \\ &= \{\omega \mid X^3(\omega) \leq c\} \in \mathcal{F} \end{aligned}$$

is an event for all $c \in \mathbb{R}$.

$$\begin{aligned}
Y^{\leftarrow}((-\infty, c]) &= \{\omega \in \Omega \mid X^3(\omega) \leq c\} \\
&= \{\omega \in \Omega \mid X(\omega) \leq \underbrace{\sqrt[3]{c}}_{c_1}\} \\
&= X^{\leftarrow}((-\infty, c_1])
\end{aligned}$$

Because X is a r.v., $X^{\leftarrow}((-\infty, c_1])$ is always an event $\in \mathcal{F}$.

Prove that $Y(\omega) = X^2(\omega)$ is a r.v. too.

Remark: The term random variable is a mishmash, because a random variable is neither random, nor is it a variable. A random variable, as we observed, is

just a function from the sample space Ω to real numbers $\mathbb{R}$.

The term "random" actually refers to the randomness in

observing a particular member
(or particular members) of a
sample space Ω , as the
outcome of an experiment.

The random variable maps

the outcome (ω) to a "fixed"
real number, $X(\omega)$.

One can imagine different
members of the sample
space are selected

randomly, and are mapped
into different real numbers,
through the random variable.

However, the random variable
does not assign real numbers

to the members of the sample
space "randomly"

Remark: It is also VERY IMPORTANT

to note that probability
is assigned to subsets

of the sample space (events)

So: $\mathbb{P}: \mathcal{F} \rightarrow \underline{[0,1]}$, but

a random variable assigns a real number to each element of the sample space, therefore

$$X: \Omega \rightarrow \mathbb{R}.$$

Borel Sets and The Probability Law of A Random Variable

Def (σ -field generated by a set)

Let C be any collection of

subsets of Ω , i.e. $C \subseteq 2^{\Omega}$, which
is not necessarily a σ -field.

Define $\sigma(C)$ to be the smallest
 σ -field that contains C , which

means that if D is a

$$C = \{A\}$$

$$\mathcal{F}_1 = \{A, A^c, \emptyset, \Omega\}$$

$$\mathcal{F}_2 = \{A, B, A^c, B^c, A \cap B, A \cup B, \dots, \emptyset, \Omega\}$$

σ -field and $C \subseteq D$, then
 $\sigma(C) \subseteq D$.

Example: We previously saw
that the smallest σ -field that
contains some $A \subseteq \Omega$ is:

$$F = \{\emptyset, A, A^c, \Omega\}.$$

F is the σ -field generated
by $C = \{A\} \subseteq 2^\Omega$.

It is obvious that any other
 σ -field that contains $C = \{A\}$

is a superset of $\sigma(C) =$
 $\{\emptyset, A, A^c, \Omega\}$. For example,
 if $B \neq A$, the σ -field that
 has both B , A as
 its members, is a superset

of $\sigma(C) = \{A, A^c, \Omega, \emptyset\}$.

Exercise: Try to enumerate the
^{minimal} members of the σ -field that has
 both A and B as its members.

$\sigma(\{A, B\})$

Borel σ -field:

Def! The Borel σ -field of subsets of $\mathbb{R}$, denoted as $\mathcal{B}(\mathbb{R})$, or simply $\mathcal{B}$, is a σ -field generated by all of the intervals

$(-\infty, c]$, $c \in \mathbb{R}$.

Question: Are intervals in the form $(c, +\infty)$, $c \in \mathbb{R}$ and (c, d) , $c, d \in \mathbb{R}$ members of $\mathcal{B}$?

$$(c, +\infty) = (-\infty, c]^c \in \mathcal{B}(\mathbb{R})$$

$$(-\infty, d) = \bigcup_{n \in \mathbb{N}} (-\infty, d - \frac{1}{n}] \in \mathcal{B}(\mathbb{R})$$

$\in \mathcal{B}(\mathbb{R})$

$$(c, d) = (-\infty, d) \cap (c, +\infty) \in \mathcal{B}(\mathbb{R})$$

It can be proved that all
intervals, their countable unions and
intersections, and any set that
 is a result of countable applications
 of $\cup, \cap$, and $\setminus$ to intervals are in $\mathcal{B}(\mathbb{R})$.

Remark: It can be proved that any singleton $\{a\}$ is a Borel Set, $\forall a \in \mathbb{R}$, e.g. $\{1\}$, $\{2.3\}$, $\{\pi\}$ are Borel Sets.

— Any finite/countably infinite

subset of $\mathbb{R}$ is a Borel Set.

— Remark: Borel Sets should not be confused with the Borel σ -field itself.

Borel Sets are the members of $\mathcal{B}(\mathbb{R})$, the Borel σ -field.

$(-\infty, c]$

- Proposition: The pre-image

$X^{-1}(B)$ of any Borel Set

$B \in \mathcal{B}(\mathbb{R})$ is an event, if

X is a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$

(Sketch of Proof): Any

Borel Set is a Countable

Union of intervals of the form

$(-\infty, c]$ or their complements

$(c, +\infty)$, so,

$$B = \bigcup_n A_n$$

where A_n is either $(-\infty, c_n]$
or $(-\infty, d_n]^c$:

$$\overset{\leftarrow}{X}(B) = \overset{\leftarrow}{X}\left(\bigcup_{n \in \mathbb{N}} A_n\right)$$

$$= \bigcup_{n \in \mathbb{N}} \underbrace{\overset{\leftarrow}{X}(A_n)}_{\in \mathcal{F}} \in \mathcal{F}$$

This is not
a proof. The
exact proof needs transfinite

$$\overset{\leftarrow}{X}(B_1 \cap B_2 \cup B_3) \stackrel{\text{induction}}{=} \underbrace{\left[\overset{\leftarrow}{X}(B_1^c) \cup \overset{\leftarrow}{X}(B_2^c) \right]}_{\cup \overset{\leftarrow}{X}(B_3)}$$

The Probability Law of A
Random Variable

Since $\forall B \in \mathcal{B}(\mathbb{R})$, $\overset{\leftarrow}{X}(B)$ is
an event (is $\mathcal{F}$ -measurable),
we can define the

probability of $X^{\leftarrow}(B)$.

(S-F,P)

$$P(X^{\leftarrow}(B)) = P(\{\omega \mid X(\omega) \in B\})$$

$$= P_X(B)$$

$P(X \in B)$

Precisely, $(\mathcal{R}, \mathcal{B}(\mathcal{R}), P_X)$ is

a probability space induced/

created by X , where

$$P_X(B) = P(X^{\leftarrow}(B))$$

$$= P(X \in B)$$

Shorthand for $X^{\leftarrow}(B)$

Example: Assume that

$$\Omega = \{1, 2, 3, 4, 5, 6\} \quad \text{and} \quad P(A) = \frac{|A|}{6}$$

$$\forall \omega \in \Omega, \quad X_1(\omega) = 2 \quad \text{and} \quad X_2(\omega) = \omega^2$$

$$X_3(\omega) = \sqrt{\omega}$$

Calculate $P_{X_1}(B)$ and $P_{X_2}(B)$

when: $P_{X_3}(B)$

$$a) B = (-\infty, -10]$$

$$P_{X_1}(B) = P(X_1(\omega) \in (-\infty, -10])$$

$$= P(\{\omega \mid 2 \in (-\infty, -10]\}) = 0$$

$$P_{X_2}(B) = P(X_2(\omega) \in (-\infty, -10])$$

$$= P(\{\omega \mid \omega^2 \in (-\infty, -10]\}) = P(\emptyset) = 0$$

$$P_{X_3}(B) = 0$$

$$(b) \quad B = [1.5, 2.5]$$

$$P_{X_1}(B) = P(\{\omega \mid X_1(\omega) \in [1.5, 2.5]\}) \\ = P(\{\omega \mid 2 \in [1.5, 2.5]\}) = P(\Omega) = 1$$

$$P_{X_2}(B) = P(\{\omega \mid \omega^2 \in [1.5, 2.5]\}) \\ = P(\emptyset) = 0$$

$$P_{X_3}(B) = P(\{\omega \mid \sqrt{\omega} \in [1.5, 2.5]\}) \\ = P(\{3, 4, 5, 6\}) = 4/6$$

$$(c) \quad B = \mathbb{N}$$

$$P_{X_3}(B) = P(\{\omega \mid X_3(\omega) \in \mathbb{N}\})$$

$$= P(\{\omega \mid \sqrt{\omega} \in \mathbb{N}\}) = P(\{1, 4\}) = 2/6 = 1/3$$

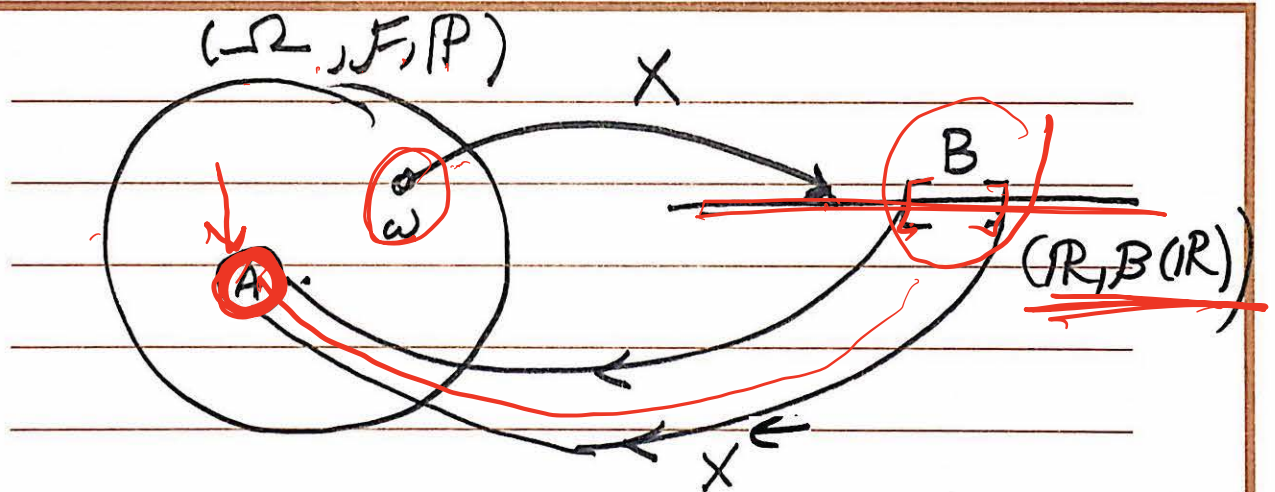
The rest are left as

an exercise.

Remark: Borel sets shed light on the usefulness of the concept of a random variable. A random variable helps creating a probability space on $\mathbb{R}$,

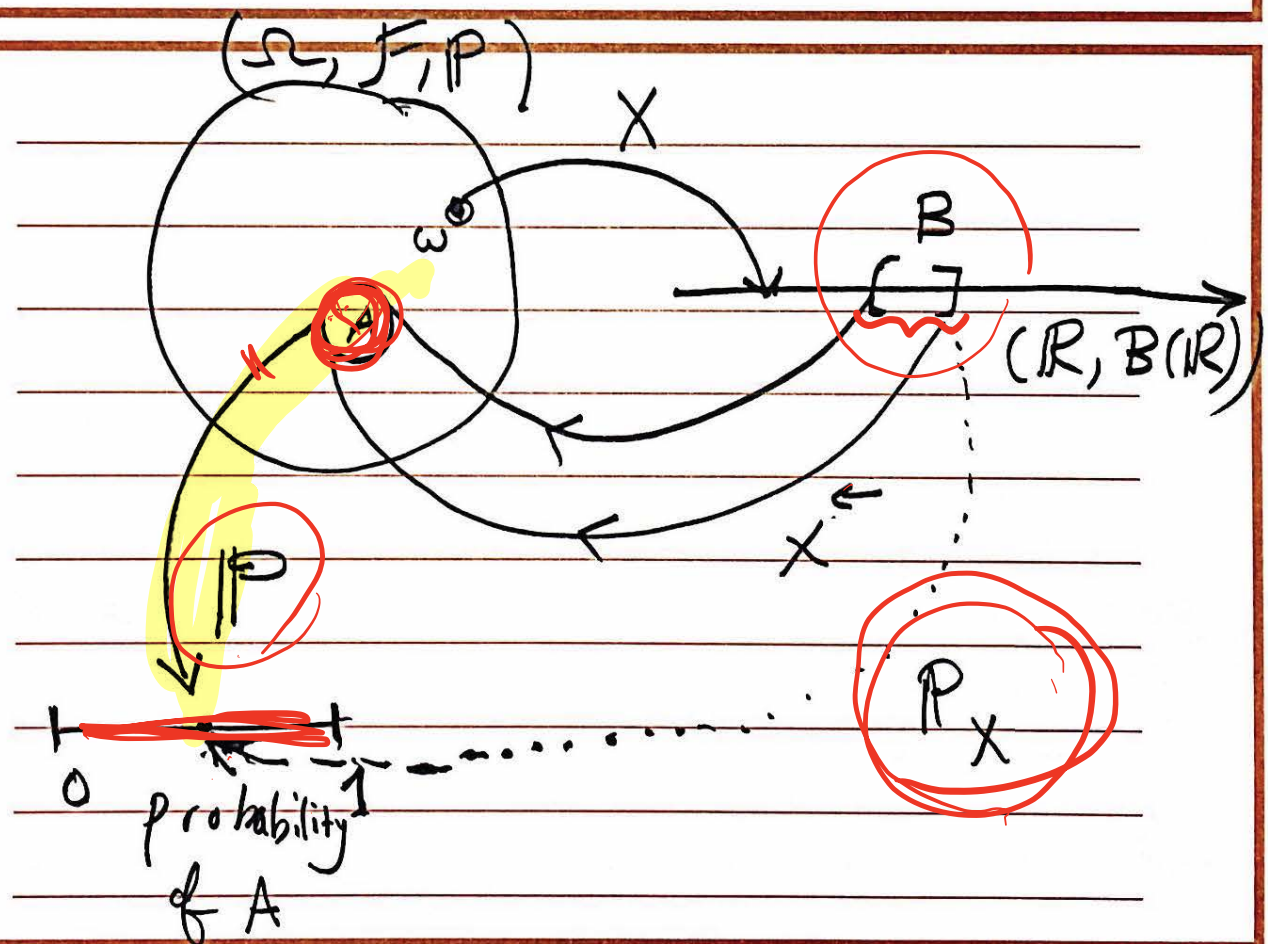
whose σ -field is $\mathcal{B}(\mathbb{R})$, and its probability measure is $P_x(\cdot)$. Instead of dealing with general sample spaces and σ -fields, by using a

random variable, we can
 deal with $\mathbb{R}$ and $\mathcal{B}(\mathbb{R})$,
 and compute probabilities
 of "numerical" measurable
 sets such as intervals.



In this picture, X is a random
 variable that maps every
 element ω in the sample

space to a real number,
 in a way that the pre-image
 of every Borel Set $B \in \mathcal{B}(\mathbb{R})$
 is an event $A \in \mathcal{F}$.



In this picture, $X^{\leftarrow}$ maps
 any Borel subset of $\mathbb{R}$ into an
 event^A, and the probability
 measure $\mathbb{P}$ maps the event
 into a number in $[0, 1]$.

Therefore, the probability law
 of X , $\mathbb{P}_X$, can be viewed as
 composition of $\mathbb{P}$ and $X^{\leftarrow}$;

$$\mathbb{P}_X(B) = \mathbb{P}(X^{\leftarrow}(B)) = \mathbb{P} \circ X^{\leftarrow}(B)$$

Theorem (Construction of r.v.'s)

(a) If $(\Omega, \mathcal{F}, P)$ is a probability space, and $A \in \mathcal{F}$ (is an event), the corresponding indicator function of A , $I_A(\omega)$ is a random

variable, where

$$I_A(\omega) = \begin{cases} 0 & \omega \notin A \\ 1 & \omega \in A \end{cases}$$

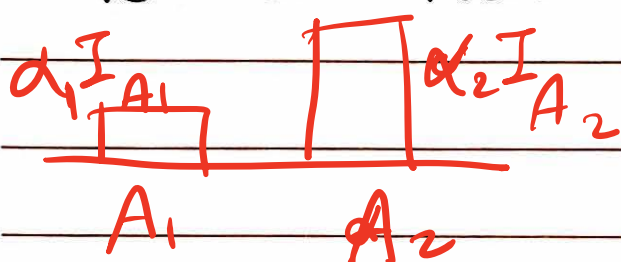
Bernoulli r.v.

(b) If $A_1, A_2, \dots, A_n \in \mathcal{F}$ and

$\alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{R}$, then:

$$X(\omega) = \sum_{i=1}^n \alpha_i I_{A_i}(\omega)$$

is a random variable.



Multinomial/
Categorical

(c) Any continuous real function

is a random variable. So, if $\Omega = \mathbb{R}$,

$X: \mathbb{R} \rightarrow \mathbb{R}$ is a random variable

if X is a continuous function, i.e.

$$\text{If } \omega \rightarrow \omega_0 \quad X(\omega) = \underline{X(\omega_0)} \quad \forall \omega_0 \in \mathbb{R}$$

(d) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function (i.e. $\lim_{x \rightarrow x_0} f(x) = f(x_0) \forall x_0 \in \mathbb{R}$) then $Y = f(X)$ is also a random variable

$$Y(\omega) = f(X(\omega)) \quad \frac{1}{X(\omega)^2 + 1}$$

Remark: (d) doesn't mean that

if f is not continuous, $f(X)$

is ~~NECESSARILY~~ not a random variable. $f(X)$ CAN ^{still} be a

random variable if f is not continuous

(e) (Functions of Multiple Random Variables) Assume that $X_1, \dots, X_n$ are random variables on $(\Omega, \mathcal{F}, P)$ and assume that $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous function.

Then, $Y = f(X_1, X_2, \dots, X_n)$ is also a random variable.

Example : $y = f(x_1, x_2) = \underline{x_1 + x_2}$

and $y = g(x_1, x_2) = \underline{x_1 x_2}$ are

continuous functions, so $X_1 + X_2$ and

X_1, X_2 are random variables

if X_1 and X_2 are random variables.

$$Y(\omega) = \min(X_1(\omega), \dots, X_n(\omega))$$

$$Z(\omega) = \max(X_1(\omega), \dots, X_n(\omega))$$

Order Statistics

The proof of some of the above statements is beyond the scope of this class.

The Cumulative Distribution Function (CDF)

Def: The cdf (cumulative distribution function) of a r.v. X , $F_X: \mathbb{R} \rightarrow [0, 1]$ is defined as:

$$F_X(x) = \mathbb{P}(X \leq x)$$

event

$$\{X \leq x\} = X^{-1}((-\infty, x])$$

$$= \{\omega \mid X(\omega) \in (-\infty, x]\}$$

Always an event by our definition
of a random variable.

Ques: Explain why any random
variable has a well-defined
CDF, i.e. $\forall X, F_X(x)$ exists
for all $x \in \mathbb{R}$.

Because we defined r.v.s
such that

$\forall x \in \mathbb{R}, X^{\leftarrow}((-\infty, x])$
is ALWAYS an event
and

$$P[X^{\leftarrow}((-\infty, x])] = F_X(x)$$

is well-defined/always exists.

Lemma : A cdf of X , a
random variable defined on
 $(\Omega, \mathcal{F}, \mathbb{P})$, has the following
properties:

$$(a) \lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow +\infty} F_x(x) = 1$$

$$(b) x < y \Rightarrow F_x(x) \leq F_x(y)$$

(It is monotonic)

F_x is non-decreasing



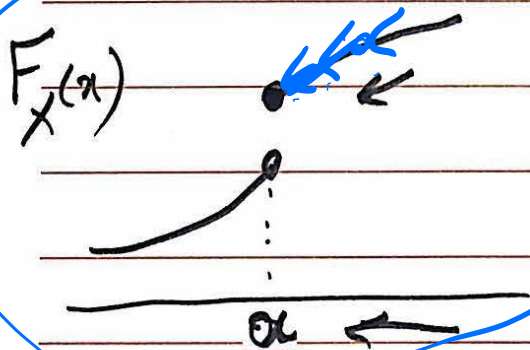
(c) F is right-continuous,

i.e.

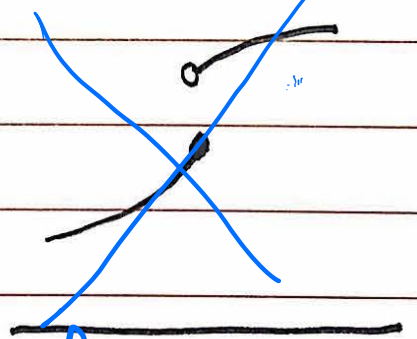
$$\lim_{x \rightarrow a^+} F_x(x) = F_x(a)$$

or

$$F(x+h) \rightarrow F(x) \text{ when } h \rightarrow 0^+$$



right-continuous



$\lim_{x \rightarrow a^-} F_x(x) \neq F_x(a)$
left-continuous

Proof: $\lim_{x \rightarrow -\infty} F_X(x) = 0$

(a) Define $B_n = \{\omega \in \Omega \mid X(\omega) \leq -n\}$
 $A_n = (-\infty, -n]$ $X^{\leftarrow}((-\infty, -n])$

$$A_1 \supseteq A_2 \supseteq A_3$$

B_n 's are decreasing

$$X^{\leftarrow}(A_1) \supseteq X^{\leftarrow}(A_2) \supseteq X^{\leftarrow}(A_3) \supseteq \dots$$

$$B_1 \supseteq B_2 \supseteq B_3 \supseteq \dots$$

$$\lim_{n \rightarrow \infty} B_n = \bigcap_{n \in \mathbb{N}} B_n = \emptyset$$

$$P(B_n) = P(\{\omega \mid X(\omega) \leq -n\})$$

$$= P(X \leq -n) = F_X(-n)$$

$$\lim_{n \rightarrow \infty} P(B_n) = \lim_{n \rightarrow \infty} F_X(-n)$$

By continuity of probability

$$P(\underbrace{\lim_{n \rightarrow \infty} B_n}_{\emptyset}) = \underbrace{\lim_{n \rightarrow \infty} F_X(-n)}_{\lim_{x \rightarrow -\infty} F_X(x)}$$

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

To prove $\lim_{x \rightarrow \infty} F_X(x)$, we similarly need to define an appropriate set of B_n 's. It is left as an exercise.

Use

$$A_n = (-\infty, n]$$

$$\lim_{n \rightarrow \infty} A_n = \mathbb{R}$$

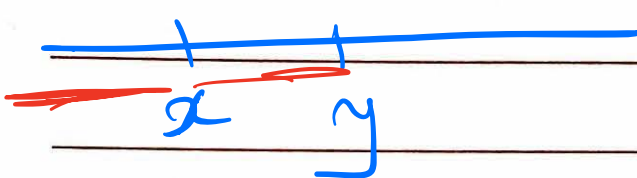
$$B_n = X^{\leftarrow}(A_n)$$

The proof will be similar and is left to the student as an exercise.

$$(b) \quad F_X(x) = P(\underbrace{X \leq x}_{\text{event}})$$

$$F_X(y) = P(\underbrace{X \leq y}_{\text{event}})$$

$$x < y$$



See the extra
page

$$\{\omega \mid X(\omega) \leq y\} = \underbrace{\{\omega \mid X(\omega) \leq x\}}_{\text{event}} \cup \underbrace{\{\omega \mid x < X(\omega) \leq y\}}_{\text{event}}$$

(C) Again, another appropriate set of events has to be defined.

Then, one should use continuity of probability.

(b) continue &

disjoint
events

$$P(X(\omega) \leq y) = P(X \leq a)$$

$$+ P(a < X(\omega) \leq y)$$

$$F_X(y) = F_X(a) + \text{non-negative}$$

$$F_X(y) \geq F_X(a)$$

$$(c) \quad B_n = \left\{ \omega \mid X(\omega) \leq a + \frac{1}{n} \right\}$$

$$A_n = (-\infty, a + \frac{1}{n}]$$

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$$

$B_n = X^{\leftarrow}(A_n)$ is also decreasing

$$B_1 \supseteq B_2 \supseteq \dots$$

$$\lim_{n \rightarrow \infty} B_n = \bigcap_{n \in \mathbb{N}} B_n$$

$$= \{\omega \mid X(\omega) \leq a\}$$

$$\lim_{n \rightarrow \infty} P(B_n) = \lim_{n \rightarrow \infty} F_X(a + \frac{1}{n})$$

By continuity of

prob

$$P(\lim_{n \rightarrow \infty} B_n) = \lim_{n \rightarrow \infty} F_X(a + \frac{1}{n})$$

$$P(\underbrace{\{\omega \mid X(\omega) \leq a\}}_{F_X(a)}) = \lim_{n \rightarrow \infty} F_X(a + \frac{1}{n})$$

$$F_X(a) = \lim_{x \rightarrow a^+} F_X(x)$$

Example (Constant Random Variable)

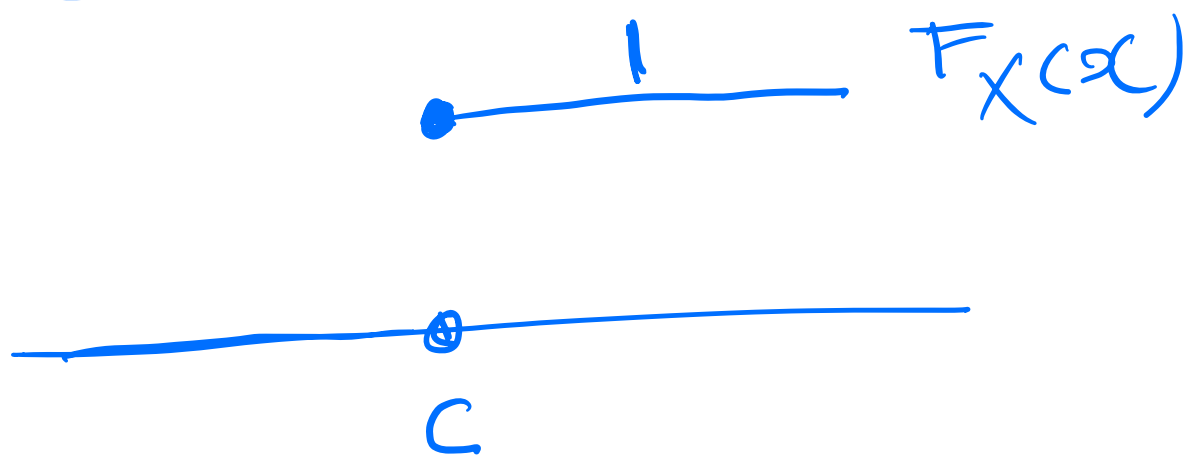
Let $c \in \mathbb{R}$, $X: \Omega \rightarrow \mathbb{R}$

$$X(\omega) = c \quad \forall \omega \in \Omega$$

What is the cdf of X ?

$$F_X(x) = P(\underbrace{X \leq x})$$

$$= \begin{cases} P(\emptyset) = 0 & x < c \\ P(\Omega) = 1 & x \geq c \end{cases}$$



It has all the properties of a cdf.

A constant r.v. is sometimes called a degenerate r.v.

Lemma & Properties of cdf.

Let X be a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$

and F_X be the cdf of X

Then:

$$(a) \mathbb{P}(X > x) = \mathbb{P}(X \in (x, +\infty))$$

Borel set

$$= 1 - \mathbb{P}(X \leq x) = 1 - F_X(x)$$

$$(b) \mathbb{P}(x < X \leq y) = \mathbb{P}(X \in (x, y])$$

Borel set

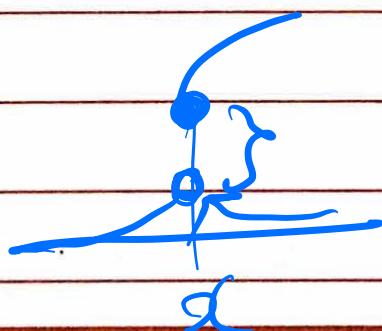
$$= F_X(y) - F_X(x)$$

$$(c) \mathbb{P}(X = x) = \mathbb{P}(X \in \{x\})$$

Borel Set

$$= F_X(x) - \lim_{y \rightarrow x^-} F_X(y)$$

$y \uparrow x$



For $\mathbb{C}$

$$B_n = \left\{ x - \frac{1}{n} < X(\omega) \leq x + \frac{1}{n} \right\}$$

or

$$B_n = \left\{ \omega \mid x - \frac{1}{n} < X(\omega) \leq x \right\}$$

Example (Constant Random
Variable)

$$X: \Omega \rightarrow \mathbb{R}$$

$$X(\omega) = c$$

What is $P(\{X=c\})$?

$$P(X=c) = F_X(c) - \lim_{x \rightarrow c^-} F_X(x)$$

$$1 - 0 = 1$$

Exercise: Assume that X is a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$ and F_X is its CDF.

$$G(x) = [1 - F(x)] \log(1 - F(x)) + F(x)$$

cdf of a random variable?

Checking the properties of a cdf will be enough

$$1. \lim_{x \rightarrow -\infty} G(x)$$

$$= \lim_{x \rightarrow -\infty} \{ [1 - F_X(x)] \log(1 - F_X(x)) + F_X(x) \} = 1 \log 1 + 0 = 0$$

$$2. \lim_{x \rightarrow +\infty} G(x)$$

$$= \lim_{x \rightarrow +\infty} [1 - F_X(x)] \log [1 - F_X(x)]$$

$0 \quad x \rightarrow +\infty \quad \rightarrow 0^+$

$$+ F_X(x) = 1$$

$$\lim_{x \rightarrow +\infty} \underbrace{[1 - F_X(x)]}_u \log \underbrace{[1 - F_X(x)]}_u$$

$$= \lim_{u \rightarrow 1^-} \underbrace{(1-u)}_z \log \underbrace{(1-u)}_z$$

$$= \lim_{z \rightarrow 0^+} \frac{z}{0} \log z = \lim_{z \rightarrow 0^+} \frac{\log z}{\frac{1}{z}}$$

L'Hôpital's rule

$$\lim_{z \rightarrow 0^+} \frac{\frac{1}{z}}{-\frac{1}{z^2}} = \lim_{z \rightarrow 0^+} -z = 0$$

We must show $G(x)$ is non-decreasing

$$G(x) = \underbrace{[1 - F_X(x)]}_{u} \log \underbrace{[1 - F_X(x)]}_u + \underbrace{F_X(x)}_u$$

$$h(u) = \underbrace{(1-u)}_u \log(1-u) + \underbrace{u}_u$$

If we prove that $h(u)$ is non-decreasing, because $F_X(x)$ is non-decreasing, then $G(x)$ is

non-decreasing

$$h'(u) = -\log(1-u) + (1-u) \frac{(A)}{1-u} + 1$$

$$= -\log(1-u) \geq 0$$

$$0 \leq u \leq 1$$

$\Rightarrow h$ is non-decreasing w.r.t. u

$\Rightarrow G(x)$ is non-decreasing
w.r.t. x

$$\lim_{x \rightarrow a^+} G(x)$$

$$= \lim_{x \rightarrow a^+} \left\{ \underbrace{(1 - F_X(x))}_{\text{}} \log \underbrace{(1 - F_X(x))}_{\text{}} + \underbrace{F_X(x)}_{\text{}} \right\}$$

$$= (1 - F_X(a)) \log(1 - F_X(a)) + F_X(a)$$

$$= G(a)$$

(because F_X is a cdf and
is right continuous and $\log$ is
continuous)

$\Rightarrow G(x)$ is a cdf.

Exercise: Are the following
statements equivalent

(1) $F_X(a)$ is continuous for
all $a \in \mathbb{R}$

(2) $P\{X=a\}=0 \quad \forall a \in \mathbb{R}$

$$P\{X=a\}=0 \quad \forall a \in \mathbb{R}$$

$$\Leftrightarrow F_X(a) - \lim_{x \rightarrow a^-} F_X(x) = 0$$

$$\Leftrightarrow \lim_{x \rightarrow a^-} F_X(x) = F_X(a)$$

On the other hand, $F_X(\cdot)$

is right continuous, so

$$\lim_{x \rightarrow a^+} F_X(x) = F_X(a)$$

$$\Leftrightarrow \lim_{x \rightarrow a^-} F_X(x) = \lim_{x \rightarrow a^+} F_X(x) \\ = F_X(a) \quad \forall a$$

$\Leftrightarrow F_X(a)$ is continuous for
all $a \in \mathbb{R}$

Moral of the story: r.v.'s
that assign zero prob to
single points have continuous
cdf's and vice versa.

Exercise: let X be a r.v. with continuous cdf F_X . Find the cdf of the following random variables.

(a) $Y = X^2$ $F_Y(y) = P(Y \leq y)$

$$= P(X^2 \leq y) = \begin{cases} 0 & y < 0 \\ P(-\sqrt{y} \leq X \leq \sqrt{y}) & y \geq 0 \end{cases}$$

$$= P(-\sqrt{y} \leq X \leq \sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

(b) $Y = \sqrt{X}$

$$F_Y(y) = P(Y \leq y) = P(\sqrt{X} \leq y)$$

$$= \begin{cases} 0 & y < 0 \\ P(0 \leq \sqrt{X} \leq y) & y \geq 0 \end{cases}$$

$$P(0 \leq \sqrt{x} \leq y) = P(0 \leq x \leq y^2) \\ = F_X(y^2) - \underbrace{F_X(0)}$$

If X is a positive random variable i.e. if $X \geq 0$ w.p.1 or a.s.

$$P(\{\omega \mid X(\omega) \geq 0\}) = 1$$

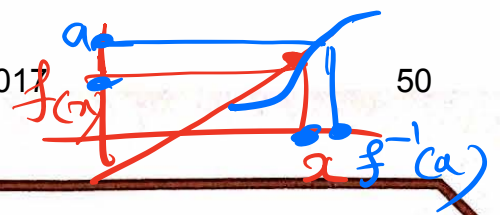
$$P(\{\omega \mid X(\omega) < 0\}) = 0$$

$$P(X < 0) = 0 \Rightarrow F_X(x) = 0 \quad x < 0$$

$$\Rightarrow \lim_{x \rightarrow 0^-} F_X(x) = 0$$

Because the cdf is continuous, it means $F_X(0)$, $F_Y(y) = F_X(y^2)$

$$f(x) < a \quad x < f^{-1}(a)$$



(c) $Y = F_X(X)$ For simplicity = assume

F_X is strictly increasing

$$\begin{aligned} F_Y(y) &= P(F_X(X) \leq y) \\ &= P(X \leq F_X^{-1}(y)) = F_X(F_X^{-1}(y)) \\ &= y \end{aligned}$$

$F_Y(y) = \int_0^y 1 \, dy$

$0 \leq y \leq 1$

$y < 0$

$y > 1$

(d) $G^{-1}(X)$, where G is a continuous and strictly increasing function

$Y = G^{-1}(X)$

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(G^{-1}(X) \leq y) \\ &= P(X \leq G(y)) = F_X(G(y)) \end{aligned}$$

Exercise:

Let X have the CDF

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}x & 0 \leq x \leq 2 \\ 1 & x > 2 \end{cases}$$

and let $Y = X^2$. Find

$$(a) P(1 \leq X < 2)$$

$$= P(1 \leq X \leq 2) = F_X(2) - F_X(1)$$

$$= \frac{1}{2}(2) - \frac{1}{2}(1) = \frac{1}{2}$$

$$= P(1 \leq X \leq 2) = P(1 < X \leq 2)$$

$$= P(1 < X < 2)$$

Remember $P(a < X \leq b) = F_X(b) - F_X(a)$

$$(b) P(2X + Y \leq 3)$$

$$P(2X + X^2 \leq 3)$$

$$= P(X^2 + 2X + 1 \leq 4)$$

$$= P((X+1)^2 \leq 4)$$

$$= P(-2 \leq X+1 \leq 2)$$

$$= P(-3 \leq X \leq 1)$$

$$= F_X(1) - F_X(-3) = \frac{1}{2} - 0 = \frac{1}{2}$$

Independence of Random Variables

Recall that the events $A, B \in \mathcal{F}$ are said to be independent iff :

$$P(A \cap B) = P(A)P(B)$$

Based on the concept of independence for events we define independence for r.v.'s :

Def: ^{Assume} X_1, X_2 are r.v.'s on $(\Omega, \mathcal{F}, P)$. They are said to be

independent, iff for all Borel

sets $B_1, B_2 \in \mathcal{B}(\mathbb{R})$:

$$P(X_1 \in B_1, X_2 \in B_2) = \overbrace{P(X_1 \in B_1)}^{\text{event}} \overbrace{P(X_2 \in B_2)}^{\text{event}}$$

$\xleftarrow{X_1(B_1)} \cap \xleftarrow{X_2(B_2)}$

~~=~~

Question: How can we extend the definition of independence a set of r.v.'s $\{X_i \mid i \in I\}$, where I is an index set?

Recall $\{A_i \mid i \in I\}$ is indep

iff $\forall J \subseteq I \quad |J| < \infty$

$$\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \mathbb{P}(A_i)$$

For r.v.'s, they are indep

iff $\forall J \subseteq I, \quad |J| < \infty$

$$\mathbb{P}\left(\bigcap_{i \in J} X_i^{\leftarrow}(B_i)\right) = \prod_{i \in J} \mathbb{P}(X_i^{\leftarrow}(B_i))$$

Where B_i 's are any Borel
Sets, i.e. $B_i \in \mathcal{B}(\mathbb{R})$.

Def(Joint cdf):

Assume that X_1 and X_2 are
r.v.'s on $(\Omega, \mathcal{F}, \mathbb{P})$. The

joint cdf of X_1 and X_2 is defined

$$\omega \quad F_{X_1, X_2}(x_1, x_2) = \mathbb{P}(X_1 \leq x_1, X_2 \leq x_2)$$

$$= \mathbb{P}\left(\underbrace{\{\omega \mid X_1(\omega) \leq x_1\}}_{X_1 \leftarrow (-\infty, x_1]} \cap \underbrace{\{\omega \mid X_2(\omega) \leq x_2\}}_{X_2 \leftarrow (-\infty, x_2]}\right)$$

More generally, if $X_1, X_2, \dots, X_n$ are random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, the joint cdf of $X_1, X_2, \dots, X_n$ is denoted as $F_{X_1, X_2, \dots, X_n}$ and is defined as:

$$\begin{aligned}
 & \mathbb{R}^n \longrightarrow \mathbb{R} \\
 & F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) \\
 &= \mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) \\
 & \quad \swarrow \quad \downarrow \quad \text{---} \\
 & \quad X_1 \in (-\infty, x_1] \quad \cap \quad \dots
 \end{aligned}$$

lemma: X_1, X_2 are
 independent if and only if
 $\forall x_1, x_2 \in \mathbb{R}$

$$F_{X_1, X_2}(x_1, x_2) = \underbrace{F_{X_1}(x_1)}_{\text{Joint}} \underbrace{F_{X_2}(x_2)}_{\text{Marginals}}$$

More generally, $X_1, X_2, \dots, X_n$ are
 independent if and only if

$$F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) =$$

$$F_{X_1}(x_1) F_{X_2}(x_2) \dots F_{X_n}(x_n)$$

Remark: The previous lemma states that instead of checking the following equation for all possible Borel sets A, B ,

$$P(X_1 \in A, X_2 \in B) = P(X_1 \in A)P(X_2 \in B)$$

one can check it only for A 's and B 's in the following

form $A = (-\infty, x_1]$, $B = (-\infty, x_2]$

$$P(X_1 \in A, X_2 \in B) = P(X_1 \in (-\infty, x_1],$$

$$X_2 \in (-\infty, x_2]) = P(X_1 \in (-\infty, x_1]) \times P(X_2 \in (-\infty, x_2])$$

$F_{X_1}(x_1)$ $F_{X_2}(x_2)$

$$\times P(X_2 \in (-\infty, x_2])$$

$F_{X_2}(x_2)$

$\{x_i \mid i \in I\}$ are independent

iff for all finite $J \subseteq I, |J| < \kappa$

$$F_{x_{i_1}, x_{i_2}, \dots, x_{i_n}}(\dots) = F_{x_{i_1}}(\dots) F_{x_{i_2}}(\dots) \dots F_{x_{i_n}}(\dots)$$

$i_1, i_2, \dots, i_n \in J$

are distinct members of J